

Oracle Flashback

Oracle flashback allows you to move database back in time.

With Flashback Database, you can rewind the database to a previous point in time without media recovery (that is, without restoring backup copies of the datafiles). When the database is running, Flashback Database buffers and writes before-images of data blocks to *flashback logs*, which are generated by default for all permanent tablespaces and which reside in the fast recovery area (FRA).

You can use flashback technology to move entire database or a particular table inside database.

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Note: only for flashback database activity, you must enable flashback database. For all other flashback activities, you do not need to enable flashback database

Flashback Table Before Drop

You can flashback a dropped table from recyclebin using flashback table command

SHOW RECYCLEBIN;

FLASHBACK TABLE "BIN\$gk3lsj/3akk5hg3j2lkl5j3d==\$0" TO BEFORE DROP;

or

FLASHBACK TABLE SCOTT.FLASH_EMP TO BEFORE DROP;

You can even rename table while flashing it back from recyclebin

FLASHBACK TABLE SCOTT.FLASH_EMP TO BEFORE DROP RENAME TO NEW_EMP;

Note: Recyclebin must be enabled to use flashback table before drop

Flashback Table

You can flashback table to a particular SCN or time in the past. Before you can flashback table, you must enable row movement

ALTER TABLE hr.employees ENABLE ROW MOVEMENT;

Now you are ready to flashback table to SCN or timestamp

FLASHBACK TABLE EMP TO SCN <scn_no>;

FLASHBACK TABLE HR.EMPLOYEES TO TIMESTAMP

TO_TIMESTAMP('2016-05-12 18:30:00', 'YYYY-MM-DD HH24:MI:SS');

Note: for flashback table, enabling FLASHBACK DATABASE is not required at all

Flashback Database

We can move an entire database back in time to a particular SCN or a timestamp. Flashback Database must be already enabled on the database to user this feature.

Enable Flashback Database

Make sure [DB_RECOVERY_FILE_DEST](#) parameter is set. This is the location where Oracle will store flashback logs

Set [DB_RECOVERY_FILE_DEST](#) parameter as per requirement

```
SQL> alter system set db_recovery_file_dest_size=50G SCOPE=spfile;
```

```
SQL> alter system set db_recovery_file_dest='/u02/flash_logs' SCOPE=spfile;
```

Confirm that the database is in ARCHIVELOG mode, which is required for Flashback Database, and enable ARCHIVELOG mode if needed.

```
ARCHIVE LOG LIST;
```

```
ALTER DATABASE ARCHIVELOG;
```

Set the [DB_FLASHBACK_RETENTION_TARGET](#) parameter which specifies the upper limit (in minutes) on how far back in time the database can be flashed back

```
SQL> alter system set db_flashback_retention_target=2880;
```

Set the associated undo retention, required for certain flashback features. Here, we set a 24-hour undo retention (in seconds), equivalent to half the DB_FLASHBACK_RETENTION_TARGET, as in above step.

```
ALTER SYSTEM SET UNDO_RETENTION=86400;
```

Enable flashback database which requires database bounce

```
SQL> shutdown immediate;
```

```
SQL> startup mount;
```

```
SQL> alter database flashback on;
```

```
SQL> alter database open;
```

```
SQL> select flashback_on from v$database;
```

Create Sample User

Let us capture the database SCN number before we create a user

```
SQL> SELECT current_scn, SYSTIMESTAMP FROM v$database;
```

Current SCN: 2703232

Create a user FLASH_USR and try to connect the database with same user

```
SQL> create user flash_usr identified by flash_usr;
```

```
SQL> grant connect, resource to flash_usr;
```

```
SQL> conn flash_usr/flash_usr;
```

Flashback Database to SCN or Timestamp

Assume that the user has been created by mistake and you want to flashback database to the SCN just before the user creation. Shutdown DB and startup mount

```
SQL> shut immediate;
```

```
SQL> startup mount;
```

Flashback database to SCN before user creation and open database with resetlogs

```
SQL> Flashback database to scn 2703232;
```

```
SQL> Alter database open resetlogs;
```

You can flashback database to particular timestamp too

```
FLASHBACK DATABASE TO TIMESTAMP
```

```
TO_TIMESTAMP('2016-05-12 18:30:00', 'YYYY-MM-DD HH24:MI:SS');
```

Reference:

<https://logicalread.com/oracle-12-flashback-database-mc07/#.YXpP3Z5BzIV>

<https://www.support.dbagenesis.com/post/oracle-flashback>